SPRINT 3 REPORT

From skill statements to leaf topics

Sprint 3 — Representation & Clustering · Section B (skills) · Taxonomy of Work capstone, Northeastern

Where Sprint 2 left us

Corpus

Kaggle LinkedIn dataset: 123,849 US postings (Mar–Apr 2024). Duplicates tagged, not dropped. Plus a live Greenhouse feed (453).

Gold set

100 hand-labelled postings, 8 sectors, 1,333 gold skills. Rule E: ESCO + O*NET decide skill vs trait.

Extractor

Qwen3.8 27B on Groq (free), prompt v3. Test F1 strict .66 / lenient .74 — ties Claude Haiku (.66 / .75).

Pipeline

Clean → dedupe → LLM extract (cached, Langfuse) → ESCO / O*NET link → Supabase. 69% known · 2% trait · 29% new.

What was still missing

- A list of skill statements, but no structure
- "excel", "microsoft excel" and "advanced excel" were three unrelated strings
- Nothing said which skills belong together, or which domains need a generic skill

Sprint 3 at a glance

Goal (brief): turn skill statements into leaf topics, with topic and membership tables and a stability report.

clean skills
(from 11,449 names)

11,093

leaf topics
HDBSCAN

374

stability AMI
on 80% subsets

0.84

library skills
249 strong links

196

- Chosen: HDBSCAN, min cluster size 10, min samples 1, on UMAP-10 of bge-small embeddings
- Every skill keeps a topic: 78% clustered, the rest are weak members of the nearest topic
- All tables, including the embeddings (pgvector), are in Supabase

What Sprint 3 asks, and our answer

Brief task	Our answer
Choose a representation and store it in the shared database	bge-small-en-v1.5 (384 dims) · Supabase pgvector table, 11,093 rows
Fit and compare at least two clustering methods	Four: HDBSCAN, Leiden, agglomerative, k-means
Run a parameter sweep and report stability	20 settings + 6 stability runs on the final data
Can a statement belong to more than one topic?	One main topic + weak members + skill-library links
Deliverable: topic and membership tables	topics (374), skill_topics (11,093) in Supabase

How we did it: eight steps

1

Snapshot

28,777 mentions · 1,931 postings

2

Clean names

11,449 → 11,093 skills

3

Embed

bge-small, 384 dims

4

Cluster

4 methods

5

Sweep

20 settings

6

Stability

80% subsets + seeds

7

Membership

main + weak + related

8

Library

196 skills, links

Snapshot = one model per posting (Qwen where its batch is done, the Haiku research run elsewhere; both score the same on test).

Cleaning skill names before clustering

Each name is one point in the embedding space, so duplicates split counts. We cross-checked every pair with independent signals:

Spelling

plural, spaces, "(ap)"
power point = powerpoint

Vendor / level word

adobe photoshop = photoshop
advanced excel = excel

Meaning + reference

near-identical embedding or same ESCO / O*NET entry

pairs merged after reading the posting evidence (of 289 checked)

269

Kept apart (different things)

- canvas (Canvas LMS) ≠ canva (design tool)
- "SAPs" (lab analysis plans) ≠ SAP software
- routings (manufacturing) ≠ routing (networks) · CMM ≠ CMMS

clean skills (from 11,449)

11,093

Representation: one vector per skill

Model

BAAI/bge-small-en-v1.5, pinned revision
384 numbers per skill, normalised (cosine = dot)
UMAP → 10 dims for HDBSCAN; 3D only for the map

Stored in the shared database

Supabase table skill_embeddings (pgvector)
11,093 rows, HNSW cosine index
Postgres similarity = Python similarity (checked)

Weakness we found: names fool the model

Skill	Lands next to
numpy	basic math, matlab
tensorflow	airflow, kubeflow ("-flow")
r	rhel, redcap
microsoft excel	microsoft office, word (not spreadsheets)

Fix planned for Sprint 4: embed the name plus a short description, e.g. the O*NET software category.

Clustering methods we tried

HDBSCAN	dense areas in UMAP space; sparse skills = noise	**374 topics · 78% · AMI 0.84**	chosen
Leiden (RB)	communities in a 15-nearest-skill graph	**323 topics · 100% · AMI 0.75**	runner-up
Leiden (CPM)	same graph, other quality function	**8 settings · Jaccard ≤ 0.53**	rejected
Agglomerative	merge closest pairs, cut at a distance	**357 topics · AMI 0.61**	hierarchy only
k-means	k round groups, fixed in advance	**cannot leave a skill out**	not kept

We tested the AI advice instead of trusting it

Leiden with CPM is steadier	Rejected	CPM Jaccard 0.53 vs RB 0.59
Merge verified aliases first	Done	269 evidence-checked merges
Never force a skill into a topic	Done	assigned / weak + score + margin
Build the hierarchy over centroids	Agreed	agglomerative least stable on raw skills
Give soft skills their own family	Done differently	the skill library + 39 transversal topics

Why we built a skill library

Clustering groups skills by meaning, from the name only. It never looks at which jobs use a skill.

1

Generic skills get stranded

HDBSCAN puts many in noise (attention to detail has no topic) or in 39 soft-skill topics, cut off from the domains that use them.

2

Our end goal needs them

"Build a finance team" needs accounting topics plus excel and attention to detail. Only the postings show that link.

3

The brief asks for it

"Decide whether a statement may belong to more than one topic." Library links are our multi-topic answer.

4

It prepares Sprint 5

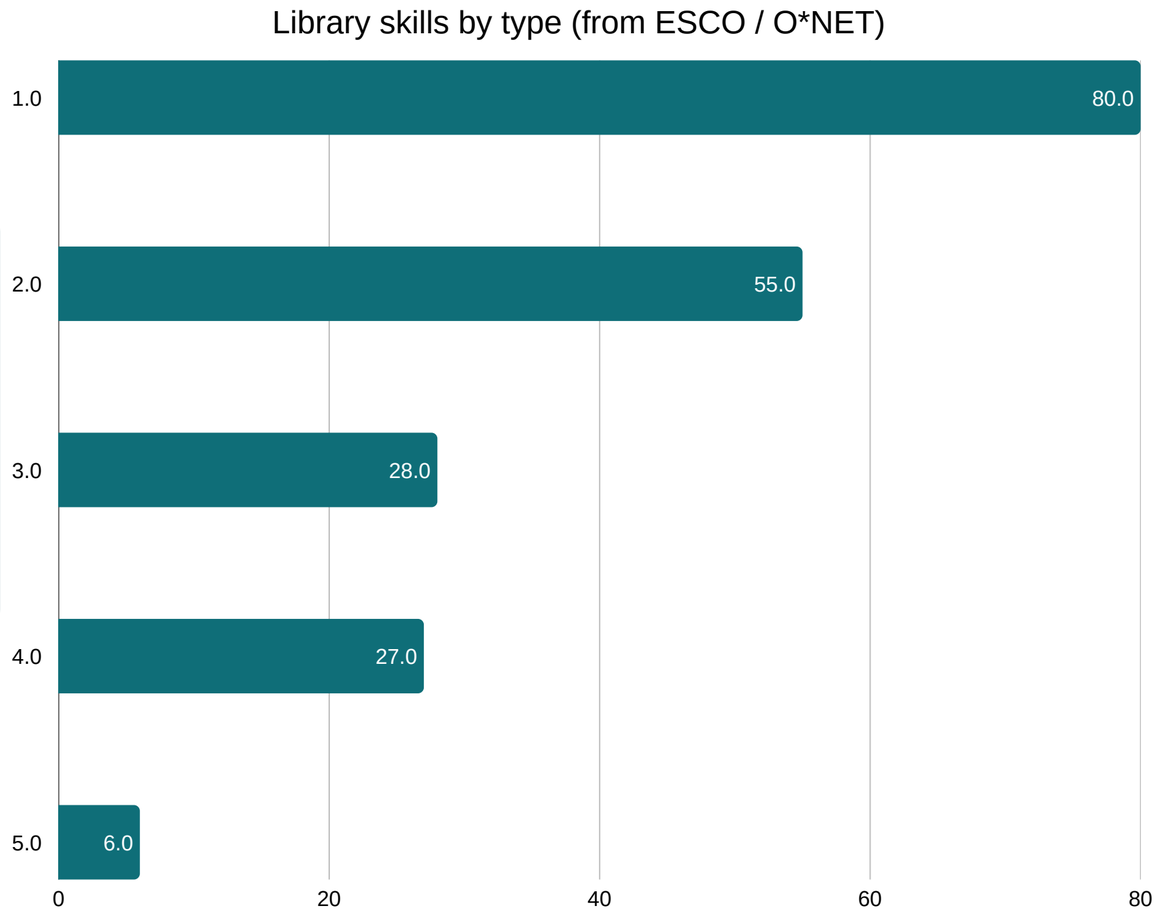
Task ↔ skill mapping with Section A uses the same co-occurrence maths: lift, p, q per job template.

The skill library: skills every sector needs

Clustering says what a skill means. Communication is in 603 postings across all 8 sectors; attention to detail has no topic at all.

Who is in the library (rule v2)

≥ 5 postings · ≥ 4 of 8 sectors · no sector > 50%
Confirmed at ≥ 10 postings · fixed before looking at results



library skills (146 confirmed)

196

of them are HDBSCAN noise

43

Linking library skills to domain topics

Example: excel and the accounts-payable topic, 1,931 postings

postings with excel	postings with the topic	shared by chance	shared in reality
191	41	≈ 4	18

- Lift = observed ÷ expected: 18 ÷ 4.1 ≈ 4.4
- p = hypergeometric test (could it be luck?)
- q = Benjamini–Hochberg over all 65,584 pairs
- Strong = ≥ 5 job templates, lift ≥ 1.5, q < 0.05

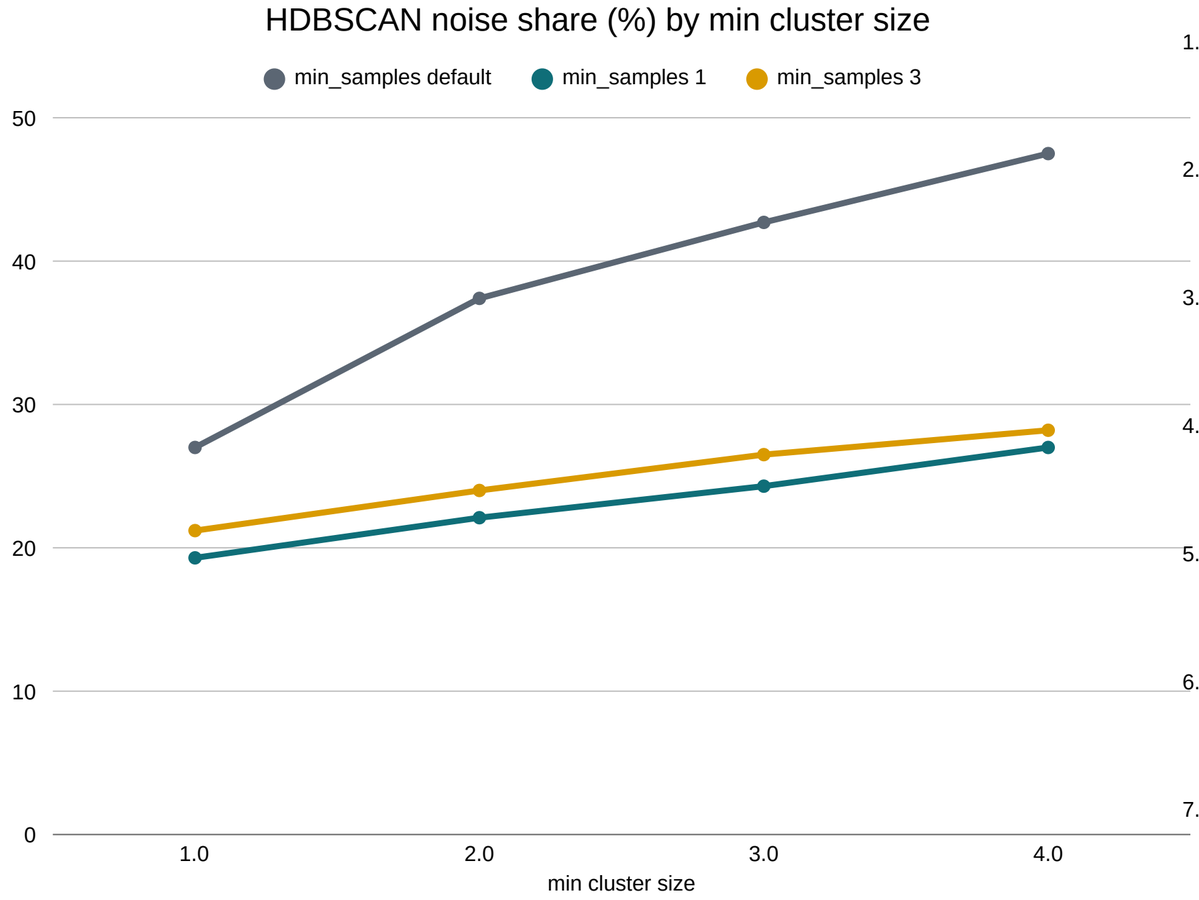
Result

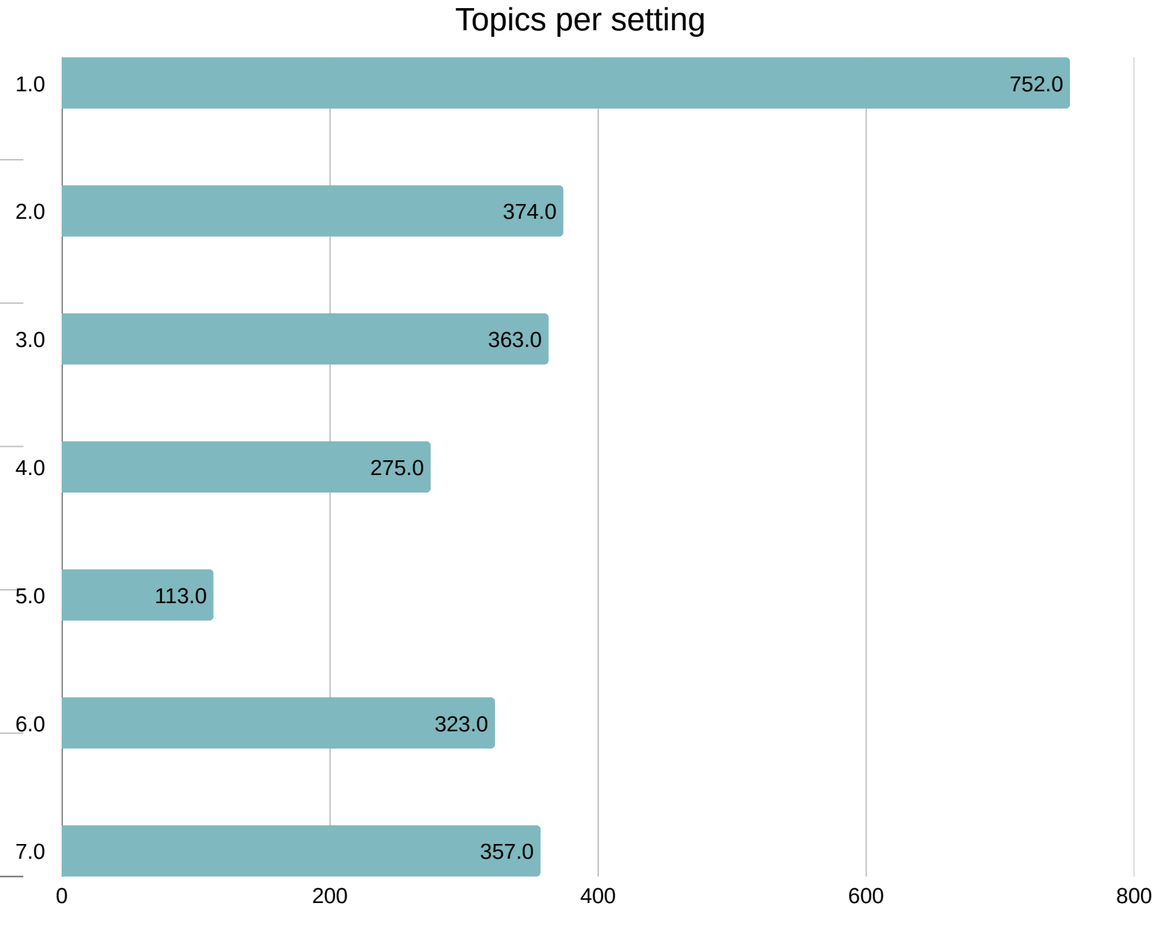
980 links kept, 249 strong
e.g. microsoft excel → payment reconciliation, lift 3.2

v1 → v2 improvements

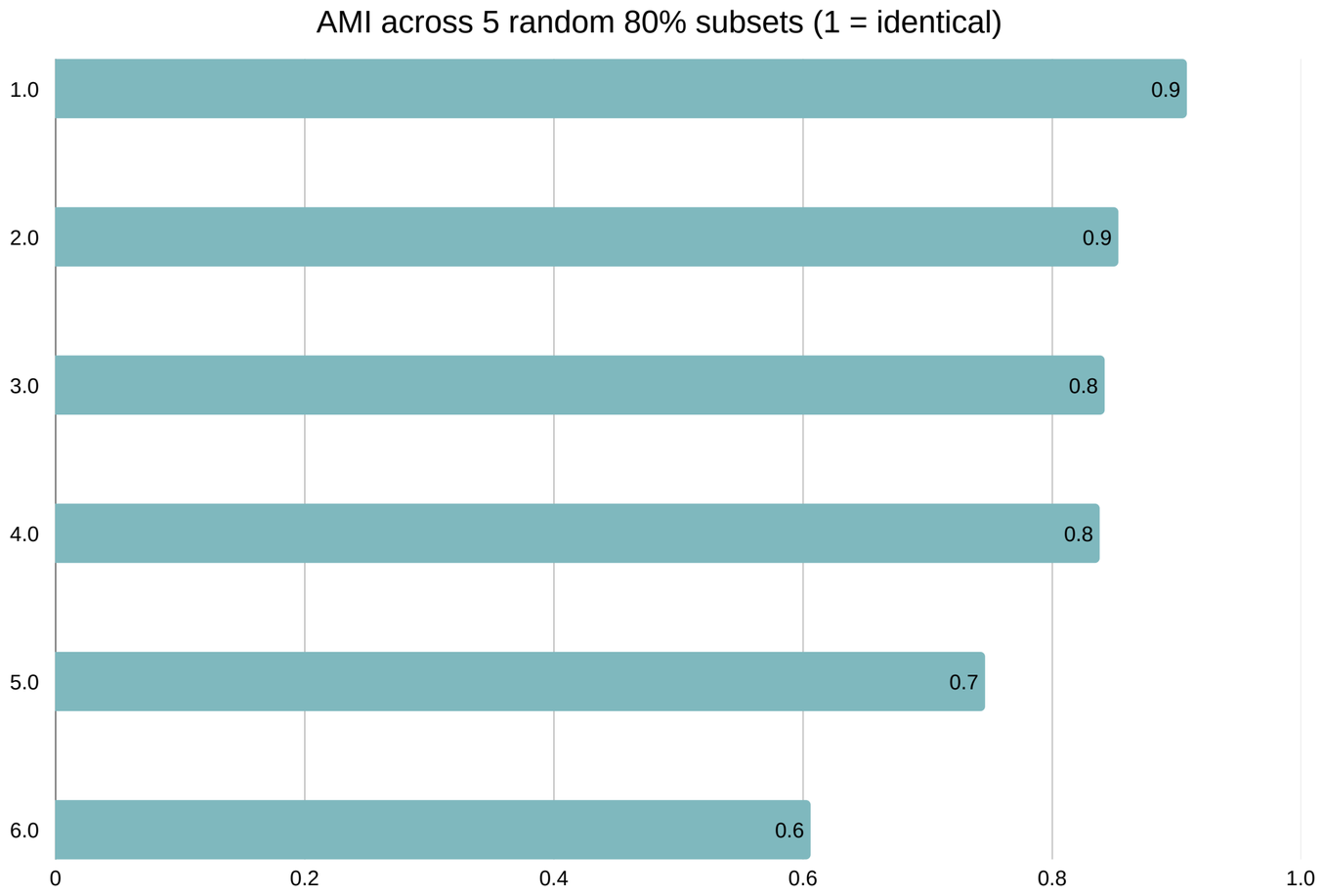
Names merged first (excel = microsoft excel)
Count per job template (company + title): 255 → 249 strong
Whole-company counting tried and rejected (staffing firms)

Parameter sweep: 20 settings on the final data





Stability: same answer on 80% subsets?



Setting	Coverage
HDB 10, default	63%
HDB 10, ms3	76%
HDB 10, ms1 (chosen)	**78%**
HDB 15, ms1	76%
Leiden res 20	100%
Agglo 0.4	100%

- "Default" looks most stable only because it drops 37% of skills
- ms3 is a near-tie; we keep ms1 for coverage
- About half of the topics dissolve in some subset, for every method

Final result of Sprint 3

leaf topics

374

skills in a topic

8,646

weak members

2,447

library links (249 strong)

980

HDBSCAN · min_cluster_size 10 · min_samples 2 · UMAP 10 dims (15 neighbours, cosine, seed 42) · bge-small-en-v1.5 · 11,093 skills

Brief deliverable	Status
Representation stored in the shared database	Done — skill_embeddings (pgvector)
At least two clustering methods compared	Done — four methods
Parameter sweep + stability report	Done — 20 settings, 6 stability runs
Multi-topic membership decided and implemented	Done — main topic + weak + library links
Topic and membership tables	Done — topics, skill_topics in Supabase

Limits and what comes next

Limits

- About half the topics are fragile, for every method
- The embedding sees only the name
- Topic labels are still the centroid skill
- Small sample (1,931 postings); thresholds set by us

Next

- Sprint 4: name every topic, build parent levels, evaluate against ESCO
- Keep the library as a cross-sector branch (as ESCO does)
- Sprint 5: task ↔ skill mapping with Section A (shared posting ids)
- Questions?

THANK YOU